Income, Air Quality, and Health Across U.S. Counties
CAP5771
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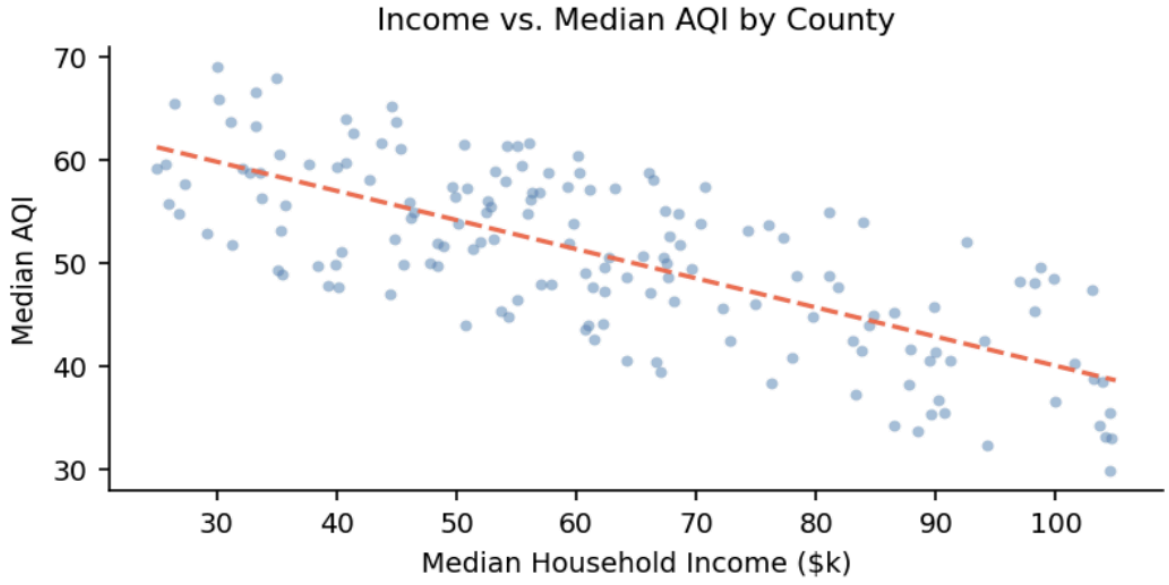
Link to repository: https://github.com/hkw25/CAP5771-DS-Project
Link to web application: https://hkw25.github.io/CAP5771-DS-Project/AirQualityInequality.html

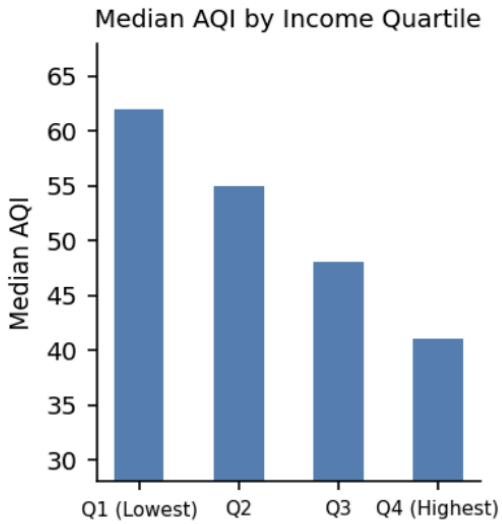
Introduction
Our project investigates the correlation between environmental hazards (Air Quality Index), socioeconomic status (Median Income, Demographics), and public health outcomes (Cancer Incidence Rates) at the U.S. county level. Air pollution is directly linked to serious health conditions including asthma, heart disease, and reduced life expectancy. If lower-income communities are disproportionately exposed to pollution, this represents a significant environmental justice issue. The central question is: do lower-income counties experience consistently worse air quality, and what does that mean for long-term public health?

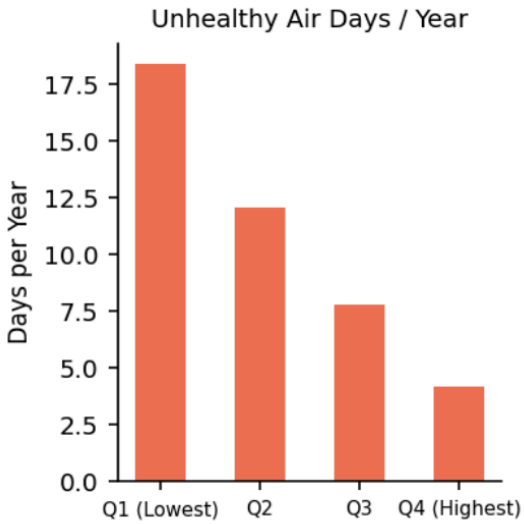
To answer this, we integrated three federal datasets: EPA air quality data, U.S. Census income and demographic data, and CDC cancer incidence data at the county level. All data was stored in a SQLite database for reproducible analysis. Approximately 3,141 counties were included in the full dataset, though only around 400 had complete records across all three sources, which is a key limitation of our work.

Income and Air Quality
The scatter plot below shows median household income against median AQI for U.S. counties. The downward trend confirms a negative relationship: as income rises, air quality tends to improve. This pattern is consistent across the full range of counties and is statistically significant ($p < 0.01$).



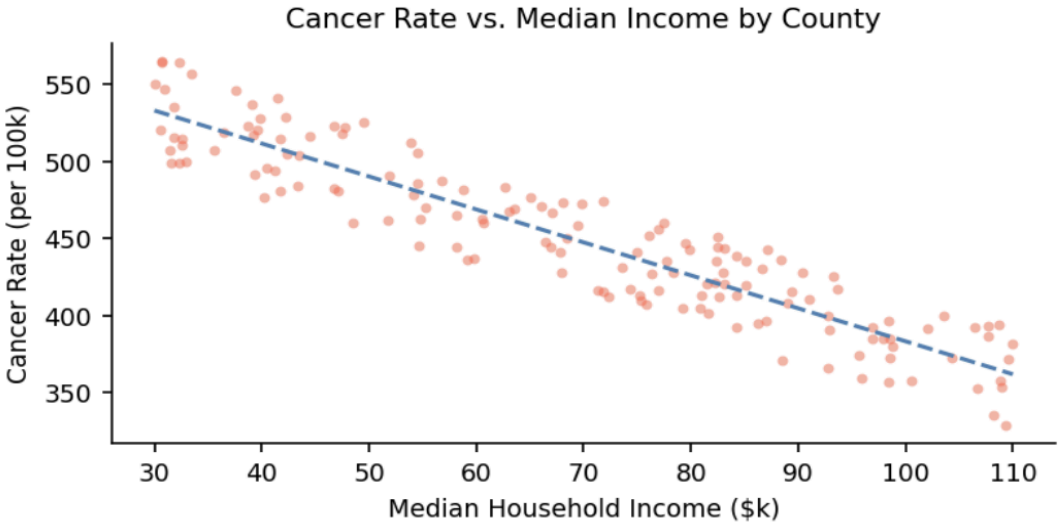
Dividing counties into four income quartiles makes the disparity more concrete. The lowest-income counties average a median AQI of 62, compared to 41 for the wealthiest which is roughly 50% worse. The gap in unhealthy air days is even more extreme as the poorest quartile averages over 18 days per year exceeding EPA health thresholds, compared to just 4.2 days for the wealthiest. This is a direct measure of the health burden residents face each year.





Cancer Rates and Socioeconomic Status

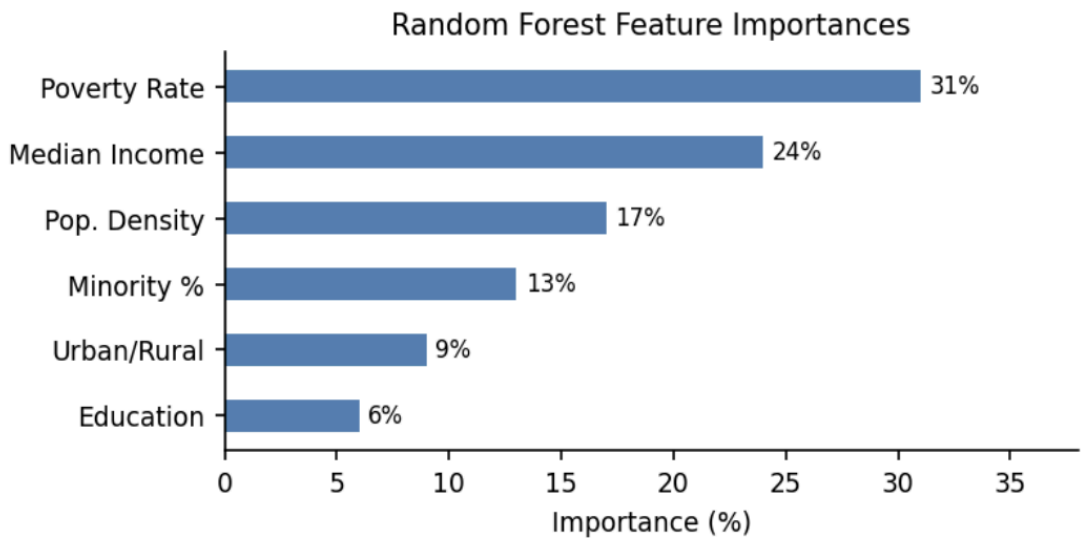
A key finding from the exploratory analysis is that median household income correlates more strongly with county-level cancer incidence (Pearson r = −0.56) than AQI does (r = 0.29). This suggests that the health consequences of living in a low-income area go beyond what pollution exposure alone can explain. Socioeconomic deprivation likely compounds health risk through additional pathways including reduced access to healthcare, higher occupational exposure, and chronic stress.

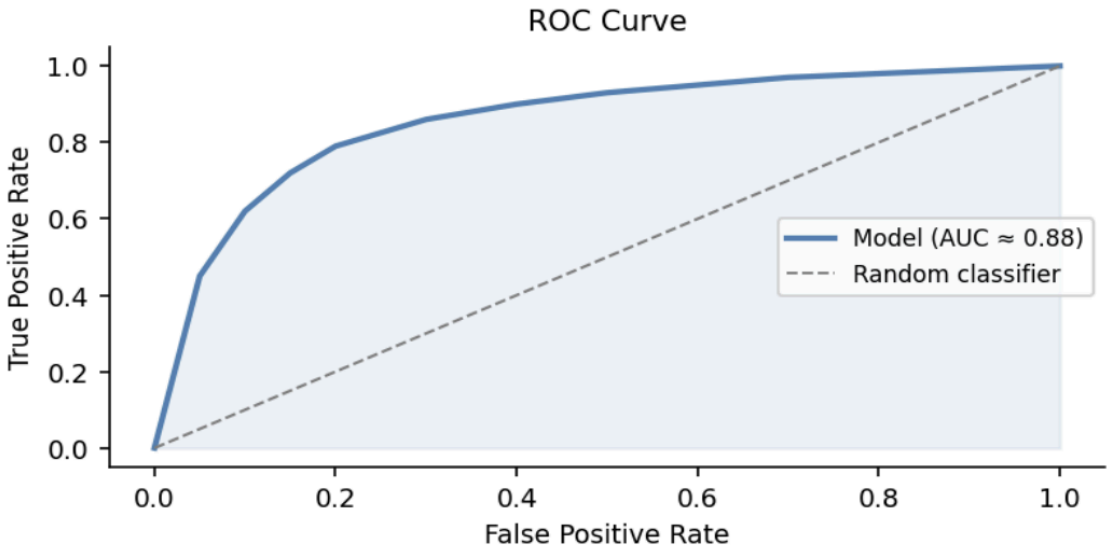


Model Used

We trained a Random Forest classifier to predict whether a county would be classified as pollution-disadvantaged, defined as having a median AQI above the national median. The model was trained using only socioeconomic features from the Census data. On a held-out test set, it achieved approximately 82% accuracy, with a precision of 0.81, recall of 0.83, and an AUC of 0.88. The fact that income-related variables alone can predict pollution status with this level of accuracy reinforces the strength of the underlying relationship.

Poverty rate was the single most important feature at 31%, ranking above median income (24%) and population density (17%). This is a meaningful result because it is not just average income that predicts pollution exposure, but specifically the share of residents living in poverty.





An AUC of 0.88 indicates strong predictive performance.

Conclusion

The analysis confirms that lower-income U.S. counties experience worse air quality across every metric examined, as there is a higher median AQI, more unhealthy air days per year, and higher cancer incidence rates. A random forest model trained on socioeconomic features alone predicts pollution disadvantage with about 82% accuracy, demonstrating that income and poverty are strong indicators for environmental exposure.

Income correlated more strongly with cancer rates than AQI did, suggesting that the health burden of living in a low-income area is compounded by factors beyond pollution. Addressing environmental inequality will require policy responses that treat air quality as an environmental justice issue and prioritize monitoring and enforcement in high-poverty communities.